

Read-only V1 API proposal

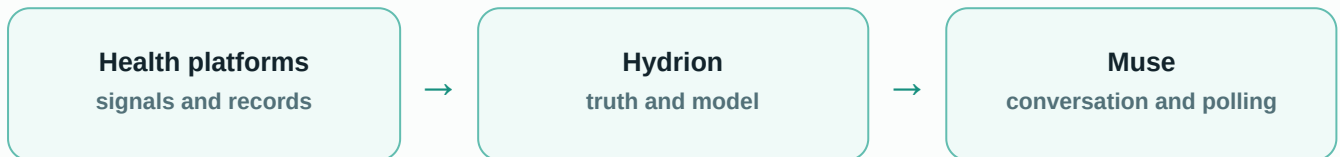
Request and response contracts, polling behavior, API-key lifecycle, and the Hydrion Settings handoff to Muse.

Review draft - for correction, not implementation lock.

Basis: Hydrion ↔ Muse Working Spec, agreed 22 September 2026.

A narrow read model preserves Hydrion as the source of truth

V1 should expose Hydrion-authored answers over public HTTPS. Muse may explain and converse about those answers, but it must not calculate hydration state, mutate logs, or alter Hydrion reminders.



Recommended V1

- Versioned REST resources under `/v1`.
- Per-user API key with explicit read scopes.
- Hydrion-calculated values plus provenance and freshness.
- Cursor-paginated, day-level history with bounded ranges.
- Conditional polling with `ETag` and `304`.
- One-time key reveal, rotation, and immediate revocation.

Explicitly not V1

- No intake creation, editing, or deletion.
- No profile, goal, medical-input, or reminder writes.
- No raw HealthKit or Health Connect relay.
- No full-database export endpoint.
- No webhook or push delivery to Muse.
- No OAuth authorization server requirement.

Core rule: if Hydrion did not return a hydration number or assessment, Muse must not invent or derive it.

Proposed API posture

HTTPS only

JSON

Read only

User scoped

Assistant agnostic

Poll friendly

The contract carries values, meaning, and freshness together

A consistent envelope lets Muse answer from Hydrion's result without guessing units, time zones, derivation, or whether the result is stale.

Transport and authorization

```
GET /v1/hydration/today
Authorization: Bearer <api_key>
Accept: application/json
X-Request-ID: <client-generated-id>
```

- TLS is required; reject plain HTTP.
- Keys identify one Hydrion user and a fixed scope set.
- Never put a key in a URL, query string, log, or response.
- Echo `X-Request-ID` when supplied; issue one otherwise.

Canonical representation

- **Volume:** integer millilitres in machine fields. Optional display fields may reflect user preference.
- **Time:** RFC 3339 timestamps with offsets; every day or period also carries an IANA time zone.
- **Dates:** ISO 8601 calendar dates interpreted in the response time zone.
- **Identifiers:** opaque strings; clients must not parse them.
- **Unknowns:** use `null` only when the field is defined but unavailable; omit optional fields that do not apply.

Standard success envelope

```
{
  "data": { "...": "resource-specific payload" },
  "meta": {
    "schema_version": "1.0",
    "generated_at": "2026-09-24T14:01:12-04:00",
    "timezone": "America/Toronto",
    "data_through": "2026-09-24T14:00:47-04:00",
    "request_id": "req_01K..."
  }
}
```

Meaning: `generated_at` is when Hydrion formed the response; `data_through` is the newest hydration state included. They are not interchangeable.

Scope vocabulary

hydration.status.read

hydration.history.read

hydration.insights.read

hydration.reports.read

hydration.reminders.read

Default recommendation: issue all five read scopes to the Muse connection, show them as grouped permissions in Settings, and reserve finer selection for a later release.

Nine resource families cover the agreed readable surface

The public connector exposes purpose-built read models, not Hydrion's tables. All endpoints are GET; the public V1 surface has no write verb.

ENDPOINT	PURPOSE	SCOPE
/v1/capabilities	Version, features, limits	Any valid key
/v1/hydration/today	Today's state and score	status.read
/v1/hydration/target	Target and derivation	status.read
/v1/hydration/days	Bounded daily history	history.read
/v1/hydration/summaries	Weekly/monthly aggregate	history.read
/v1/challenges/active	Streak and challenge state	status.read
/v1/reminders	Hydrion reminder state	reminders.read
/v1/insights	Hydrion-authored insights	insights.read
/v1/reports	Report index and detail	reports.read

Capabilities response

```
{
  "data": {
    "api_version": "v1",
    "schema_version": "1.0",
    "features": ["today", "target_derivation", "daily_history",
      "summaries", "challenges", "reminders", "insights", "reports"],
    "limits": {
      "history_max_days_per_request": 31,
      "history_oldest_available_date": "2026-01-08",
      "page_size_default": 14,
      "page_size_max": 31
    }
  },
  "meta": { "generated_at": "...", "timezone": "America/Toronto",
    "data_through": "...", "request_id": "req_01K..." }
}
```

Path versioning

Breaking contract changes get a new major path. Additive fields may appear within V1; clients ignore unknown fields.

Content versioning

`schema_version` starts at 1.0. Hydrion documents additions and deprecation windows.

Naming choice for review: endpoint nouns use general resource names while scope strings keep the `hydration.` namespace. This keeps URLs short without weakening authorization clarity.

One status response answers the common questions without recalculation

Request: GET /v1/hydration/today?date=2026-09-24. The optional date defaults to “today” in the user's Hydrion time zone and may only be today; historical dates belong to /days.

```
{
  "data": {
    "date": "2026-09-24",
    "state": "behind",
    "consumed_ml": 1250,
    "target_ml": 2500,
    "remaining_ml": 1250,
    "progress_percent": 50,
    "last_log_at": "2026-09-24T13:42:00-04:00",
    "score": { "value": 63, "scale_min": 0, "scale_max": 100,
      "label": "needs_attention", "model_version": "hydrion-score-1" },
    "display": { "volume_unit": "ml", "consumed": "1,250 ml",
      "target": "2,500 ml", "remaining": "1,250 ml" }
  },
  "meta": { "schema_version": "1.0", "generated_at": "2026-09-24T14:01:12-04:00",
    "timezone": "America/Toronto", "data_through": "2026-09-24T13:42:00-04:00",
    "request_id": "req_01K..." }
}
```

Hydrion-owned

state, remaining_ml, progress_percent, and the entire score object are returned, not derived by Muse.

Stable state enum

Proposed values: no_data, behind, on_track, met, exceeded.

Zero handling

If no intake exists, return consumed_ml: 0. Use last_log_at: null; do not omit the resource.

Target derivation is a separate explainable resource

Request: GET /v1/hydration/target?date=2026-09-24

```
{
  "data": {
    "date": "2026-09-24", "target_ml": 2500,
    "derivation": {
      "method": "baseline_plus_adjustments",
      "baseline": { "amount_ml": 2200, "basis": "profile_baseline" },
      "adjustments": [
        { "type": "weather", "amount_ml": 300,
          "label": "Warm-weather adjustment", "effective": true }
      ],
      "chosen_goal": null,
      "explanation": "Your 2,200 ml baseline includes a 300 ml warm-weather adjustment."
    }
  },
  "meta": { "schema_version": "1.0", "generated_at": "...",
    "timezone": "America/Toronto", "data_through": "...", "request_id": "..." }
}
```

Day-level history is useful without becoming a raw export

Return Hydrion's daily read model, not individual intake rows. A 31-day request ceiling and opaque cursors bound every call.

Request

```
GET /v1/hydration/days?start=2026-09-01&end=2026-09-24&limit=14
GET /v1/hydration/days?cursor=cur_eyJkYXRlIjo1...&limit=14
```

PARAMETER	RULE	DEFAULT
start, end	Inclusive dates; maximum span 31 days	Last 7 days
limit	Integer 1-31	14
cursor	Opaque; cannot mix with start/end	None

Response

```
{
  "data": {
    "days": [
      { "date": "2026-09-24", "state": "behind", "consumed_ml": 1250,
        "target_ml": 2500, "remaining_ml": 1250, "progress_percent": 50,
        "score": { "value": 63, "label": "needs_attention" },
        "log_count": 5, "first_log_at": "2026-09-24T08:04:00-04:00",
        "last_log_at": "2026-09-24T13:42:00-04:00" },
      { "date": "2026-09-23", "state": "met", "consumed_ml": 2600,
        "target_ml": 2500, "remaining_ml": 0, "progress_percent": 104,
        "score": { "value": 91, "label": "excellent" },
        "log_count": 8, "first_log_at": "2026-09-23T07:51:00-04:00",
        "last_log_at": "2026-09-23T21:10:00-04:00" }
    ]
  },
  "pagination": { "limit": 14, "next_cursor": "cur_eyJkYXRlIjo1...",
    "has_more": true },
  "meta": { "schema_version": "1.0", "generated_at": "...",
    "timezone": "America/Toronto", "data_through": "...", "request_id": "..." }
}
```

Privacy boundary: V1 deliberately omits per-log amounts, container identifiers, raw sensor records, profile inputs, and medical context. A later need for log-level history should require a separate scope and review.

Ordering and empty periods

- Sort descending by date by default. The cursor preserves the original range and order.
- Include days with zero intake only when Hydrion has a target or state for that date; otherwise omit them.
- If no days match, return days: `[]` with `has_more: false`, not 404.

Hydrion returns finished aggregates, not ingredients for a second model

Request: GET /v1/hydration/summaries?period=week&anchor=2026-09-24. Allowed periods are week and month. Hydrion defines period boundaries in the user's time zone and returns them explicitly.

```
{
  "data": {
    "period": { "type": "week", "start": "2026-09-21", "end": "2026-09-27",
      "complete": false },
    "summary": {
      "days_with_data": 4,
      "total_consumed_ml": 8650,
      "average_consumed_ml": 2163,
      "average_target_ml": 2500,
      "days_target_met": 2,
      "target_met_rate_percent": 50,
      "average_score": { "value": 78, "label": "good" },
      "best_day": { "date": "2026-09-23", "score": 91 },
      "trend": { "direction": "improving", "comparison_period": "previous_week",
        "consumed_change_percent": 8, "score_change": 5 },
      "assessment": "Hydration improved from last week, with two target days so far."
    }
  },
  "meta": { "schema_version": "1.0", "generated_at": "...",
    "timezone": "America/Toronto", "data_through": "...", "request_id": "..." }
}
```

Incomplete periods stay explicit

`complete: false` prevents Muse from presenting a partial week or month as final. The denominator for averages is `days_with_data`, calculated by Hydrion.

Trend is optional, not inferred

If Hydrion lacks a valid comparison period, return `trend: null`. Muse must not compare period totals itself.

Score contract

FIELD	PURPOSE	RULE
value	Hydrion score	Number inside declared scale
label	Stable machine meaning	Documented enum
model_version	Interpretation trace	Present on today's score
assessment	User-ready explanation	Hydrion-authored text

Review choice: the example includes Hydrion-authored trend and assessment fields because they reduce assistant-side inference. Remove either if the current domain model cannot support it reliably.

Read-only state supports conversation without shadow behavior

Challenge and streak state

GET /v1/challenges/active

```
{
  "data": {
    "streak": {
      "current_days": 6,
      "best_days": 14,
      "qualified_today": false,
      "qualification_rule_label": "Meet your daily target"
    },
    "challenges": [
      { "id": "chg_01K...", "title": "Seven-day steady hydration",
        "status": "active", "starts_on": "2026-09-20",
        "ends_on": "2026-09-26", "progress": 5,
        "goal": 7, "unit": "days",
        "progress_percent": 71 }
    ], "meta": { "...": "standard fields" }
  }
}
```

Reminder state

GET /v1/reminders

```
{
  "data": {
    "enabled": true,
    "timezone": "America/Toronto",
    "quiet_hours": { "start": "22:00", "end": "07:00" },
    "next_scheduled_at": "2026-09-24T15:00:00-04:00",
    "rules": [
      { "id": "rem_01K...", "type": "interval",
        "enabled": true, "interval_minutes": 120,
        "window": { "start": "09:00", "end": "21:00" } }
    ], "meta": { "...": "standard fields" }
  }
}
```

No parallel reminder system: this endpoint explains Hydrion's local notifications. Muse may schedule a conversational poll requested by the user, but it never writes these rules or treats `next_scheduled_at` as an instruction to send a duplicate alert.

Hydrion-generated insights

GET /v1/insights?status=active&limit=10

```
{
  "data": {
    "insights": [
      { "id": "ins_01K...", "kind": "pattern", "severity": "info",
        "title": "Afternoon intake has improved",
        "body": "Your average intake between noon and 5 pm is higher than last week.",
        "period": { "start": "2026-09-17", "end": "2026-09-23" },
        "created_at": "2026-09-24T07:10:00-04:00", "expires_at": null }
    ],
    "pagination": { "limit": 10, "next_cursor": null, "has_more": false },
    "meta": { "...": "standard fields" }
  }
}
```

Proposed enums: insight `kind` = pattern, progress, caution, milestone; `severity` = info, attention. Medical diagnosis or emergency language stays outside this connector contract.

Metadata-first reports expose Hydrion's conclusions, not a bulk export

The list endpoint is paginated and lightweight. Detail returns the report Hydrion already generated as structured JSON; binary download is deferred unless a concrete use case requires it.

List reports

GET /v1/reports?type=weekly&limit=10&cursor=...

```
{
  "data": {
    "reports": [
      {
        "id": "rpt_01K...", "type": "weekly", "title": "Hydration week",
        "period": { "start": "2026-09-14", "end": "2026-09-20" },
        "status": "ready", "created_at": "2026-09-21T07:02:00-04:00",
        "content_type": "application/json"
      }
    ]
  },
  "pagination": { "limit": 10, "next_cursor": null, "has_more": false },
  "meta": { "...": "standard fields" }
}
```

Fetch one Hydrion-generated report

GET /v1/reports/rpt_01K...

```
{
  "data": {
    "id": "rpt_01K...", "type": "weekly", "title": "Hydration week",
    "period": { "start": "2026-09-14", "end": "2026-09-20" },
    "created_at": "2026-09-21T07:02:00-04:00",
    "overview": "You met your target on five of seven days.",
    "metrics": {
      "total_consumed_ml": 16900, "average_consumed_ml": 2414,
      "average_target_ml": 2500, "days_target_met": 5,
      "average_score": { "value": 84, "label": "good" }
    },
    "sections": [
      { "heading": "What went well",
        "body": "Morning consistency improved across the week." },
      { "heading": "Opportunity",
        "body": "Two lower-intake days were concentrated late in the week." }
    ]
  },
  "meta": { "...": "standard fields" }
}
```

404 semantics

Use 404 `report_not_found` for an unknown or inaccessible identifier. Do not reveal whether another user's report exists.

Retention semantics

`capabilities.limits` should disclose the oldest available report date if report retention is bounded.

Review choice: V1 JSON-only is the safer default. If existing Hydrion reports are PDFs, add `download_url` only as a short-lived authenticated URL and define its expiry explicitly.

Predictable failures and validators make polling safe to operate

Standard error envelope

```
{
  "error": {
    "code": "invalid_date_range",
    "message": "The requested range exceeds 31 days.",
    "details": { "max_days": 31 }
  },
  "meta": { "request_id": "req_01K...", "generated_at": "2026-09-24T14:01:12-04:00" }
}
```

STATUS	STABLE CODE	CLIENT BEHAVIOR
400	invalid_request	Correct parameters; do not retry unchanged
401	invalid_api_key	Reconnect or replace key
403	insufficient_scope	Explain missing permission
404	resource_not_found	Do not retry unknown ID
422	invalid_date_range	Reduce range to declared limit
429	rate_limited	Honor <code>Retry-After</code>
503	temporarily_unavailable	Back off; report unavailable, not zero

Conditional polling contract

```
GET /v1/hydration/today
If-None-Match: "hydrion-today-u_01K-v42"

HTTP/1.1 304 Not Modified
ETag: "hydrion-today-u_01K-v42"
Cache-Control: private, max-age=60
```

- Return an `ETag` on all pollable resources. A `304` has no body and does not advance `data_through`.
- Use `Cache-Control: private`; never permit shared caches to store user hydration data.
- Rate-limit per key and endpoint family. Publish limits through response headers and capabilities once operational values are known.
- Treat `401` after prior success as likely revocation and stop polling until the user reconnects.

Starting cadence for review: poll every 15 minutes only during a user-requested check-in window; poll no more than hourly for general status refresh; stop during quiet hours. Use conditional requests. This is an operational default, not an API guarantee.

Suggested headers: `X-RateLimit-Limit`, `X-RateLimit-Remaining`, `X-RateLimit-Reset`, and `Retry-After` on 429. Final numbers should be set after server cost testing.

One-time reveal and visible revocation are enough for a practical V1

Credential model

- One personal key per connected service, labeled “Muse” by the user-facing flow.
- Opaque format with a recognizable non-secret prefix, for example `hyd_v1_...`
- At least 256 bits of randomness from a cryptographically secure generator.
- Hydrion stores only a strong hash plus metadata; the full value is shown once.
- Muse stores the key in Secure Vault and attaches it at request time.

Stored metadata

```
{
  "id": "key_01K...",
  "service": "muse",
  "label": "Muse",
  "prefix": "hyd_v1_abcd",
  "scopes": ["hydration.status.read", "..."],
  "created_at": "2026-09-24T14:20:00-04:00",
  "last_used_at": "2026-09-24T14:31:08-04:00",
  "last_used_ip": null,
  "revoked_at": null
}
```

Recommendation: do not display or retain IP addresses in the app UI for V1. A coarse “last used” timestamp is enough for user control without exposing extra network data.

Lifecycle states

1. Create. User reviews readable categories and confirms key creation.

2. Reveal once. Show the full key with Copy and Continue. Leaving the screen makes it unrecoverable.

3. Use. Hydrion shows Created, Last used, and Active. The key value never appears again.

4. Rotate. Confirming rotation invalidates the old key immediately and reveals a replacement once.

5. Revoke. Immediate, irreversible invalidation. Subsequent connector calls return `401 invalid_api_key`.

Server-side checks

- Constant-time hash comparison.
- Scope check on every request.
- Per-key rate limits and abuse monitoring.
- Redaction in access, error, and analytics logs.
- Audit events for create, rotate, revoke, and failed authentication.

Rotation trade-off: immediate invalidation is simple and safest, but it creates a short disconnect while the replacement is pasted into Muse. A grace-period rotation can be added later if this becomes painful.

The screen explains the boundary before it creates a credential

Location: **Settings** → **Connected services** → **Muse**. The first screen should say exactly what Muse can read and what it cannot change.

Connected services **Muse**

Connect Muse

Ask Muse about the hydration data and insights Hydrion has already calculated.

Hydrion stays in control. Muse reads selected Hydrion data. It cannot add drinks, change goals, edit reminders, or access raw health-platform data.

Muse can read

Today's hydration status and target	Included
Recent daily history and summaries	Included
Score, streaks, challenges, and insights	Included
Reminder state and Hydrion reports	Included
Add, edit, or delete hydration logs	Not allowed
Change profile, goals, or reminders	Not allowed

Hydrion will create a personal API key for this connection. You will copy it into Muse's secure connection screen. The key is shown once and can be revoked here at any time.

Create connection keyCancel

Confirmation dialog

Create a read-only key for Muse?

This key gives access to the categories listed above until you revoke or rotate it. Hydrion will never show the full key again.

Create keyGo back

Interaction notes

- Do not say “Connect” before the key reaches Muse; at this point Hydrion has only created a credential.
- If device authentication is available, require it before revealing a new or rotated key.
- Keep permissions grouped in V1. Granular switches imply combinations the connector may not be tested to support.

The handoff is explicit, and the steady state remains inspectable

One-time key reveal

Muse connection key

Copy this key now

You will not be able to view it again.

hyd_v1_.....

Copy key

Next

1. Open Muse's secure connection card.

2. Paste the key when Muse asks.

3. Return here to check "Last used."

I've saved the key

The masked string above is illustrative UI, not a real credential. The production screen shows the full value only until the user leaves.

Active credential view

Connected services Muse

StatusActive

Keyhyd_v1_abcd...

Created24 Sep 2026, 2:20 pm

Last used24 Sep 2026, 2:31 pm

Read access

Status and target; history and summaries; score, challenges and insights; reminder state; reports.

Rotate key

Revoke access

State and copy rules

OBSERVED STATE	STATUS LABEL	PRIMARY ACTION
Key exists, never used	Key created - not yet used	Connection instructions
Recent successful request	Active	View access
Key exists, use is old	Active - last used [time]	Troubleshoot
Revoked	Disconnected	Create new key

Avoid false certainty: Hydrion can prove that the key was used, not that Muse is continuously “connected.” The UI should prefer “Active” plus “Last used” over a live online indicator.

Seven choices need correction or acceptance before implementation

These are the proposal's deliberate defaults. Changing them does not reopen the settled architecture: V1 remains read-only, Hydrion remains authoritative, and Muse remains a polling client.

#	RECOMMENDED CHOICE	ALTERNATIVE TO CONSIDER
1	Day-level history only	Add scoped log-level history
2	31-day range; cursor pages	Smaller or larger hard ceiling
3	Five bundled read scopes	Single all-read scope
4	JSON reports in V1	Short-lived PDF download
5	Immediate key rotation	Brief old-key grace period
6	15 min check-in / 60 min general	Different cost-latency balance
7	Grouped permissions in Settings	User-selectable categories

Implementation acceptance criteria

API

- Every public V1 route rejects write verbs.
- Every hydration value is produced by Hydrion.
- All responses carry time zone and freshness metadata.
- History limits are enforced server-side.
- Conditional requests return correct validators.
- Revoked keys fail immediately.

Settings

- Readable categories and prohibited actions are explicit.
- Full key is visible once and copyable.
- Returning to the screen shows metadata only.
- Last-used wording does not imply live connectivity.
- Rotation and revocation require confirmation.
- Device authentication protects key reveal when available.

Suggested build order

1. **Freeze enums and primitives:** time, units, states, score labels, errors, and pagination.
2. **Ship the smallest useful slice:** capabilities, today, target, authentication, and Settings key flow.
3. **Add context:** days and summaries, with rate limits and conditional polling.
4. **Add authored outputs:** challenges, reminders, insights, and reports.
5. **Conformance test:** verify scopes, revocation, empty states, time-zone boundaries, partial periods, and stale-data behavior.

Proposed lock point: approve the resource map, canonical fields, scope model, range limit, report format, and key lifecycle before naming implementation classes or persistence tables.